

Assignment 5 Solutions

$$5.22 \quad \begin{bmatrix} 1 & 1 & 2 \\ 2^{-2} & 3^{-2} & 5^{-4} \\ -1^{+1} & -3^{+1} & -3^{+2} \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & -2^{+2} & -1^{+2} \end{bmatrix} \sim \begin{bmatrix} \boxed{1} & 1 & 2 \\ 0 & \boxed{1} & 1 \\ 0 & 0 & \boxed{1} \end{bmatrix}$$

Both 1-1 and onto. 1 pivot per. col and row

$$5.23 \quad \begin{bmatrix} 5 & 0 & 5 \\ 0 & 0 & 0 \\ 4 & 0 & 7 \end{bmatrix}$$

Neither. Columns are not L.I., at most 2 pivots per row.

5.25

$$\begin{bmatrix} \boxed{1} \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

1-1, not onto. 1 pivot and 1 column
but 6 rows

$$5.27 \quad \begin{bmatrix} 1 & 2 & -1 & 6 & 9 \\ 2^{-2} & 5^{-4} & -5^{+2} & 13^{-12} & 22^{-18} \\ -3^{+3} & -4^{+6} & -3^{-3} & -16^{+18} & -19^{+27} \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -1 & 6 & 9 \\ 0 & 1 & -3 & 1 & 4 \\ 0 & 2 & -6 & 2 & 8 \end{bmatrix} \sim \begin{bmatrix} \boxed{1} & 2 & -1 & 6 & 9 \\ 0 & \boxed{1} & -3 & 1 & 4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Neither. 2 pivots, 3 rows and 5 columns

6.2

$$2 \begin{bmatrix} 6 & 1 \\ -1 & -3 \\ 8 & 1 \end{bmatrix} - 4 \begin{bmatrix} 6 & -5 \\ -6 & -2 \\ -3 & 4 \end{bmatrix} - 3 \begin{bmatrix} -4 & 0 \\ -8 & 5 \\ -4 & -1 \end{bmatrix} =$$

$$\begin{bmatrix} 12 & 2 \\ -2 & -6 \\ 16 & 2 \end{bmatrix} - \begin{bmatrix} 24 & -20 \\ -24 & -8 \\ -12 & 16 \end{bmatrix} - \begin{bmatrix} -12 & 0 \\ -24 & 15 \\ -12 & -3 \end{bmatrix} =$$

$$\begin{bmatrix} (12-24+12) & (2+20) \\ (-2+24+24) & (-6+8-15) \\ (16+12+12) & (2-16+3) \end{bmatrix} = \begin{bmatrix} 0 & 22 \\ 46 & -13 \\ 40 & -11 \end{bmatrix}$$

6.5

$$\begin{bmatrix} -2 & 6 \\ -2 & 7 \\ -3 & -2 \end{bmatrix} \begin{bmatrix} -4 & 8 & 4 \\ -10 & -4 & 9 \end{bmatrix} = \begin{bmatrix} (-2(-4)+6(-10)) & (-2(8)+6(-4)) & (-2(4)+6(9)) \\ (-2(-4)+7(-10)) & (-2(8)+7(-4)) & (-2(4)+7(9)) \\ (-3(-4)+(-2)(-10)) & (-3(8)+(-2)(-4)) & (-3(4)+(-2)(9)) \end{bmatrix}$$

$$= \begin{bmatrix} -52 & -40 & 46 \\ -62 & -44 & 55 \\ 32 & -16 & -30 \end{bmatrix}$$

6.6

$$\begin{bmatrix} 8 & 6 & -1 & -2 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \\ 1 \\ 5 \end{bmatrix} = 8(3) + 6(5) - 1(1) - 2(5) =$$

$$24 + 30 - 1 - 10 =$$

43

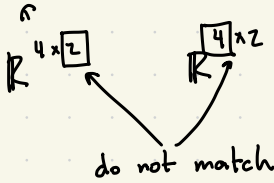
6.7

$$\begin{bmatrix} -3 \\ 6 \\ 8 \\ 5 \end{bmatrix} \begin{bmatrix} -9 & -7 & 0 & -8 \end{bmatrix} = \begin{bmatrix} 27 & 21 & 0 & 24 \\ -54 & -42 & 0 & -48 \\ -72 & -56 & 0 & -64 \\ -45 & -35 & 0 & -40 \end{bmatrix}$$

6.8

$$\begin{bmatrix} 7 & -10 \\ -6 & 6 \\ 1 & 0 \\ -4 & 2 \end{bmatrix} \begin{bmatrix} -8 & 9 \\ -10 & 0 \\ -9 & 1 \\ -4 & -3 \end{bmatrix}$$

Not well-defined



6.18

$$\begin{bmatrix} \cos 2 & -\sin 2 & 0 \\ \sin 2 & \cos 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}^{2023}$$

$$= \begin{bmatrix} \cos 4046 & -\sin 4046 & 0 \\ \sin 4046 & \cos 4046 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

This is equivalent to 2023 rotations around the z-axis by 2 rad.

5.37 True. It's the only way to have 1 pivot per column and row

5.38 True. A has a column with no pivots.

5.39 False. $\vec{x} \mapsto \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \vec{x}$

6.33 True. Stated in lecture, matrix addition is elementwise.

6.36 False. $A = B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

6.42 False.

$$\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$$

$\neq$

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix}$$

6.47 False.

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

6.49 False.

$$\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

5.57 a)

$$S(\vec{x}) = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 1 & -1 \end{bmatrix} \vec{x} \quad T(\vec{x}) = \begin{bmatrix} 1 & 1 & 1 \\ -1 & -2 & -3 \end{bmatrix}$$

$$S \circ T(\vec{x}) = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ -1 & -2 & -3 \end{bmatrix} \vec{x}$$

$$= \begin{bmatrix} 1 & 1 & 1 \\ -2 & -4 & -6 \\ 2 & 3 & 4 \end{bmatrix} \vec{x}$$

$$\begin{aligned} \text{b)} \quad & \begin{bmatrix} 1 & 1 & 1 & 1 \\ -2 & -4 & -6 & 8 \\ 2 & 3 & 4 & -3 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & -2 & -4 & 10 \\ 0 & 1 & 2 & -5 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & -5 \\ 0 & 0 & 0 & 0 \end{bmatrix} \\ & \sim \begin{bmatrix} 1 & 0 & -1 & 6 \\ 0 & 1 & 2 & -5 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{aligned}$$

$$x_1 = 6 + x_3$$

$$x_2 = -5 - 2x_3$$

x_3 is free

$$6.70 \quad \begin{bmatrix} 1 & -1 & 2 & 1 \\ -3 & 4 & 2 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 2 & 1 \\ 0 & 1 & 8 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 10 & 4 \\ 0 & 1 & 8 & 3 \end{bmatrix}$$

$$\vec{b}_1 = \begin{bmatrix} 4 \\ 3 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 & 2 & 0 \\ -3 & 4 & 2 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 2 & 0 \\ 0 & 1 & 8 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 10 & 1 \\ 0 & 1 & 8 & 1 \end{bmatrix}$$

$$\vec{b}_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$B = \begin{bmatrix} 4 & 1 \\ 3 & 1 \\ 0 & 0 \end{bmatrix}$$

* there are multiple possible answers

6.72

$$B \begin{bmatrix} 1 & -1 & 2 \\ -3 & 4 & 2 \end{bmatrix} = \left[B \begin{bmatrix} 1 \\ -3 \end{bmatrix} \quad B \begin{bmatrix} -1 \\ 4 \end{bmatrix} \quad B \begin{bmatrix} 2 \\ 2 \end{bmatrix} \right]$$

$$B = [\vec{b}_1 \quad \vec{b}_2] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \in \text{span} \{ \vec{b}_1, \vec{b}_2 \}$$

$$\Rightarrow \text{span} \{ \vec{b}_1, \vec{b}_2 \} = \mathbb{R}^3$$

This is not possible.

$$\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \in \text{span} \{ \vec{b}_1, \vec{b}_2 \}$$

$$\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \in \text{span} \{ \vec{b}_1, \vec{b}_2 \}$$

$$6.72 \text{ a)} \begin{bmatrix} 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ -4 \\ 4 \end{bmatrix}$$

$$= 3 + 4 + 4 = 11$$

$$b) \begin{bmatrix} 1 & -1 & 1 & 1 \\ -1 & 2 & -1 & 0 \\ 1 & -2 & 1 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & -1 & 0 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned} x_1 &= 2 - x_3 \\ x_2 &= 1 \\ x_3 &\text{ is free} \end{aligned}$$

$$c) \vec{v} = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 2 \end{bmatrix} A \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} =$$

$$\begin{bmatrix} 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = 0$$